AirTon project report

Specification, recovered progress and startup baseline

As of 4 October 2026 | Repository baseline v0.2.0 | Working review document

1. Product direction and present position

The founder has selected **home intraocular pressure (IOP) monitoring for patients already under a doctor's care** as AirTon's first commercial direction. The system is intended to help an enrolled patient collect eye-specific measurements at home and share traceable history with an assigned care team. Clinical design inputs, eligibility, measurement schedules and escalation instructions still need review.

The original 2024 idea submission already targeted a home-use non-contact tonometer. Later decks broadened the narrative to glaucoma screening, bilateral measurement and AI. This baseline restores a focused first indication while preserving the historical work. Pressure alone cannot establish or exclude glaucoma; retinal classification is a separate research task. [S01, S02, P02]

Actual historical project files have now been recovered: three training notebooks, six Hugging Face app snapshots, one React web application, two published Keras model files, five device drawing PDFs, pitches, a budget and IP drafting material. These establish real software experiments and concept development. The reviewed evidence does not establish a calibrated AirTon instrument, authorized home-use study, manufacturing release, patent filing, incorporation or commercial launch.

This report distinguishes source records, logged experiments, new repository work and proposed development. Source edit dates do not prove physical milestones. There is no single defensible percentage complete across all workstreams.

The detailed product specification (../docs/product/specification.md), verified status (../docs/project/status.md), evidence register (../docs/project/evidence-register.md) and source register (../research/source-register.md) are maintained alongside this dated snapshot. Original document contents are evidence, not instructions to contact people, execute notebook code, submit forms or publish claims.

2. Recovered history and configuration

Artifact date	Primary record	What the record establishes
23 August 2024	SIH2024 idea submission, P02	Problem 1550: portable non-contact home eye-pressure measurement; concept, not an award or completed instrument
November 2024-April 2025	AIRTON_ML notebook, P03	Keras CNN source and logged image-classification training
February-March 2025	Six Hugging Face Spaces, P06	Static, Streamlit, Flask/FastAPI and Node prototype sources at fixed revisions
April-May 2025	Original/copy DeepLearn notebooks, P04-P05	ORIGA-derived retinal experiments, recorded logs and figures
June-July 2025	Budget Sheet and supplied deck, P09/P01	Historical financial assumptions, block diagram, simulation screenshots and broad product claims
19 August 2025	Later pitch, P08	Arduino/ESP32 description, reported INR 7,000 prototype and projected schedule; no reconciled as-built evidence
6 September 2025	Patent-form drafting guide, P10	Preparatory material that acknowledges missing technical detail; no filing acknowledgement
4 November 2025	airton-web-final GitHub source, P07	React application at commit 062fa7ac95a13953b2c736f6b19a92dc502161bd
20 November 2025	Thiruvizhi model record, P12	Related same-author retinal research; direct AirTon linkage unconfirmed
13 September 2026	Five device drawing PDFs, P11	Concept geometry and patent-style reference views; no editable CAD or manufacturing definition
4 October 2026	Founder direction and repository work	Home monitoring priority, source recovery, audits, configuration and current report

Provider-created/modified timestamps establish artifact history. Pitch timelines projected development through July 2026; they are not substituted for completion records. Later private work may exist. The searches and reviewed connected records do not amount to an exhaustive internet inventory.

The July deck describes a Raspberry Pi 4 host, while the August pitch names Arduino/ESP32. The repository records both descriptions and keeps the actual host, acquisition controller and as-built configuration unresolved. No fabricated PCB, firmware, CAD dimensions, calibration coefficients or pneumatic limits fill these gaps. See history (../docs/project/history.md) and mechanical design history (../hardware/mechanical/design-history.md).

3. Home monitoring specification and hardware

The proposed workflow starts with clinic enrolment, patient training, a named care team, an assigned device and a clinician-defined plan. The patient confirms identity and eye; the instrument checks calibration, battery, sensing and alignment. Failed checks prevent acquisition. Unsuitable acquisitions produce a reason and no pressure value. Accepted observations retain device, firmware, calibration, quality and timestamp provenance before local storage and upload.

An encrypted durable outbox must tolerate lost connectivity and interrupted power. The server acknowledges durable receipt using immutable measurement IDs. The patient can distinguish saved locally, delivered and reviewed states. Clinician review uses eye-specific trends and the complete failure history; amendments preserve original records. The system cannot infer that an upload has been read.

An independent controller must own stimulus timing, watchdogs, exposure limits and safe shutdown. A Linux host, phone, cloud service or model cannot bypass those controls. Engineering should first establish one measurement channel and a repeatable observable on model-eye fixtures, then evaluate bilateral options.

The recovered drawings show visual device concepts, not nozzle geometry, tolerances, optical paths, material specifications, airflow waveforms, calibration or physical performance. Air-path pressure is not automatically IOP. Barometric sensor specifications do not validate tonometry. Corneal biomechanics and optical timing require characterization before selecting an estimator. [S04-S07, S31]

Unresolved design input	Required evidence
Observable, sensing rate and timing	Raw synchronized bench traces and uncertainty budget
Stimulus, optical exposure and safe failure	Reviewed limits, independent controls and fault verification
Measurement range, bias and repeatability	Comparator protocol, acceptance criteria and analysis
Home positioning and supported users	Representative critical-task studies, training and failure recovery
Form factor, battery, cleaning and service	Native design files, environmental verification and service records
Cost and production configuration	Supplier quotes, as-built BOM, calibration labour, yield and revision control

No clinical or actuation numerical limit is invented. The home configuration (../config/home-monitoring.json) deliberately leaves them null. See the optical feasibility plan (../docs/hardware/optical-feasibility.md), bench plan (../docs/hardware/bench-plan.md) and home validation plan (../docs/clinical/home-monitoring-validation.md).

4. Models, datasets actually evidenced and results

The recovered classifiers operate on retinal images. They do not estimate IOP from air-puff or optical signals. No paired AirTon/reference-pressure dataset was recovered. The model registry (../models/model-registry.json) has 11 entries, including historical, related, synthetic and planned models; it does not imply 11 trained or clinically evaluated AirTon models. The dataset registry (../data/dataset-registry.json) distinguishes actual notebook evidence from candidate releases and unverified folder/card references.

The early notebook defines a 256px Keras CNN with two convolution/pooling blocks, a dense head and sigmoid output. Logs report 455 training and 64 validation images and final validation accuracy 0.7656. Dataset origin is unresolved. Source includes an invalid expression, a zero-valued validation-step calculation, partial evaluation logic, inconsistent normalization and an incorrect equality check for sigmoid classification. These logs are historical evidence, not reproducible independent performance. [P03]

The DeepLearn notebooks download the Kaggle glaucoma-detection release containing ORIGA-derived images. Recorded distribution: 650 images, 482 normal and 168 glaucoma. The experiment samples 168 normal images to form a balanced 336-image cohort, then uses an unstratified 75/25 development/validation split, corresponding to 252/84 images. No patient-group split, frozen identifiers or independent test cohort is established. ORIGA's original publication is distinct from the mirror's rights declaration; the exact release and original rights require review. [P04-P05, S32-S33]

Experiment	Recorded selected validation result	Material limitation
VGG19, pretrained	62/84 = 73.81%; selected epoch 16	Inference cell uses epoch 11; reused selection validation
ResNet18, pretrained	62/84 = 73.81%; selected epoch 7	Freeze helper re-enables parameters; no reserved test
DenseNet169, custom head	73.81%; selected epoch 27	Sigmoid output supplied to CrossEntropyLoss
Custom VGG19 head	42.86%; selected epoch 2	Sigmoid/CrossEntropyLoss mismatch
VGG16	Architecture displayed	No evidenced training result

Manual segmentation masks are used to crop the optic disc, but the deployment path does not reproduce that input pipeline. Training normalization and inference normalization differ. A copied cup-to-disc function returns a fixed 0.6 ratio; it is a placeholder, not segmentation or measured CDR.

5. Weight integrity, result audit and data governance

Two published HDF5 weight files have been downloaded to ignored local storage. Provider SHA-256 values matched the complete downloaded bytes. HDF5 model configuration was inspected without Keras deserialization or inference.

Registry entry	Local artifact	Inspected structure / disposition
M01	Gmodel.h5; 189,104,112 bytes	Input 256 x 256 x 3; final single sigmoid output; exact training lineage unproven
M02	my_model2.h5; 74,157,296 bytes	Input 100 x 100 x 3; final single sigmoid output; exact training lineage unproven
M08	Thiruvizhi Hub metadata/code	Card claims EfficientNet/ACRIMA; config and model.py conflict; ViT-named weight not downloaded
M09	AIRTON Hub model repository	Repository exists; no model weight in the inspected revision

Full hashes, immutable revisions and retrieval commands are in the registry. `scripts/fetch_published_models.py` verifies local files by default and requires an explicit fetch selection for downloading. It never executes a model. Artifact integrity establishes identity, not dataset rights, training lineage or clinical suitability. Model licences and class mappings remain unresolved. [P06, P12]

The aggregate VGG19 and ResNet18 confusion plots label the classes in the opposite order to numeric labels used in the source. Interpreting numeric label 0 as normal and 1 as glaucoma gives VGG19 TN/FP/FN/TP = 39/9/13/23 and ResNet18 = 35/13/9/27. The resulting audit calculations are sensitivity 63.89% / specificity 81.25% for VGG19 and 75.00% / 72.92% for ResNet18. Both give 73.81% accuracy on the same 84-image selection-validation cohort. These are recalculations from recorded figures, not new experiments or clinical performance estimates.

ACRIMA's provider record identifies version 1 and CC BY 4.0. Its original paper describes 705 images: 309 normal and 396 glaucoma. It is mentioned in the related Thiruvizhi card but checkpoint use is not verified. Drishti-GS-named files are present in connected Drive metadata; their use by an AirTon checkpoint is not established. REFUGE, REFUGE2 and PAPILA remain candidates in the registry. [S10-S13, S34-S35]

No raw retinal images or patient records were imported into Git. Three tracked notebooks retain code with outputs removed; originals containing embedded images and private Drive listings remain under ignored local-private/. Only aggregate audit figures are retained. All real-data work needs rights, governance, identity and duplicate review, patient-group partitions, frozen preprocessing, a reserved external test and a controlled model card. [S15-S17]

Read the full model audit (../ml/recovered-model-audit.md) and historical model card (../models/historical-model-card.md).

6. Software, IoT, cloud and repository configuration

The actual React 19 / Vite / TypeScript app is preserved at its fixed GitHub commit under legacy/github/airton-web-final. The editable working copy is apps/web. It has patient, doctor and admin views, mock service data and a role-selection login. Random IOP values and financial figures are fictional. There is no authenticated clinical backend, device integration or verified care-team workflow. [P07]

The working copy now states demonstration status on every route, reflects the home-monitoring direction, removes unsupported accuracy/testimonial/company and HIPAA claims, restores the missing stylesheet, binds local development/preview to loopback, and removes the Vite definition that would expose a Gemini API key. Original upstream files remain unchanged. Styling now builds locally with pinned Tailwind/PostCSS dependencies and system fonts after a CDN failure was observed. Optional icons and placeholder images remain external prototype dependencies.

Six recovered Hugging Face apps document real historical implementation attempts. They are not interchangeable production services. Material issues include random disease scores in the Node API, hardcoded secret placeholders, unrestricted uploads, fragile JSON user storage, missing/import-conflicting dependencies, preprocessing mismatch and missing referenced checkpoints. Runtime metadata is a dated Hub snapshot, not end-to-end service verification. No patient data was uploaded to any public demo. See legacy audit (../docs/software/legacy-audit.md).

The separate Python reference workflow is synthetic-only. It provides strict contracts, quality/clock rejection, null IOP on rejected acquisitions, raw-file hashes, immutable idempotent SQLite ingestion, loopback bearer-token HTTP examples, HTML reports and illustrative FHIR observations. Its artificial regression exercise demonstrates subject-group splits and experiment provenance; scores have no clinical meaning. The React app is not wired to this API.

Repository area	Purpose
legacy/, ml/notebooks/archive/, artifacts/	Preserved sources, sanitized notebooks, drawings and integrity manifests
apps/web/, src/airton/, tests/, scripts/	Runnable prototype/reference software and verification tools
hardware/, firmware/, config/, schemas/	Engineering work areas and explicit current configuration boundaries
docs/, quality/, business/, reports/	Specification, decisions, clinical/quality/startup records and dated reports
data/, models/, research/	Dataset/model lineage, historical results and primary-source citations

Production work requires unique device identities, secure provisioning, patient and care-team authorization, encrypted storage/transport, durable acknowledgements, clock trust, audit trails, backup/restore evidence and incident operations. FHIR shapes are examples until a profile and interoperability testing are completed. [S26-S29] See cloud operations (../docs/cloud/operations.md).

7. Clinical, regulatory and quality pathway

The first evidence programme needs to answer three separate questions: does the instrument measure a defensible physical observable, does the complete calibrated device agree with an appropriate reference, and can intended patients operate it at home safely and produce usable records? Office accuracy alone does not answer the home-use question. FDA home-use guidance is a design reference; it does not authorize use in India. [S36]

Begin with controlled model-eye and fault verification. An authorized supervised comparator study then needs frozen design, eligibility, reference method, clinically reviewed acceptance criteria, masking/order controls, repeatability, subgroup analysis, adverse-event management and all failure/rejection records. Paired and repeated observations need appropriate clustered analysis. Sample size must follow precision and risk requirements, not a convenient round number.

Home validation adds low-vision, dexterity, language, caregiver, training-retention, alignment, storage, cleaning, network and power interruption tasks. A patient can complete a session while delivery fails; those outcomes must remain distinguishable. Clinical review assignment and support capacity are parts of the system.

CDSCO's ophthalmology list places an ophthalmic tonometer in Class B. AirTon's final indication and added software claims require qualified classification and pathway review under the Medical Devices Rules and applicable amendments. There is no recovered AirTon authorization. ISO 8612 tonometer requirements, ISO 15004-2:2024 optical hazard protection, ISO 14971:2019 risk management and IEC 62304 software lifecycle are relevant reference areas. Catalogues establish scope; licensed normative clauses and applicability assessment are still needed. [S18-S20, S23-S25, S37]

India's final DPDP Rules were notified in 2025 with staged commencement. The official rule, timeline and corrigendum must be read together at the time of each processing/release decision; this repository does not treat all obligations as already commenced on 4 October 2026. Data notices, purpose, rights handling, vendor responsibilities and security controls need reviewed implementation. [S21-S22]

The repository now provides a design record index, risk and requirements records, review/change/CAPA concepts, design-history structure, production-master structure and per-device history template. These are organized working records, not an ISO certificate or approved manufacturing file. Supplier qualification, incoming inspection, production tests, calibration traceability, serial/lot history, complaints, service and surveillance remain release work. See quality system (../docs/quality/quality-system.md) and post-market/service plan (../docs/quality/post-market-and-service.md).

8. Business, affordability and intellectual property

The pitches describe a low-cost product, but prototype price and a business forecast are not an as-built cost or sales ledger. The reported INR 7,000 prototype has no itemized supporting BOM. Other capital/prototype estimates differ across documents. No incorporation, financing receipt, commercial invoice or sales record was recovered. These states remain unverified. [P01, P08-P09]

The financial Sheet uses INR 35,000 selling price, INR 25,000 COGS and INR 12,59,000 annual fixed costs. Its monthly unit rows sum to 200; the displayed annual total says 180. Both are preserved as historical planning scenarios, recalculated by scripts/financial_scenarios.py.

Reconciled hardware-only scenario	Stated 180 units	Row-derived 200 units
Revenue	INR 63,00,000	INR 70,00,000
COGS	INR 45,00,000	INR 50,00,000
Contribution	INR 18,00,000	INR 20,00,000
Result after listed fixed costs	INR 5,41,000	INR 7,41,000

Contribution is INR 10,000 per unit, or 28.57% of selling price; hardware-only break-even is 126 units using these assumptions. Consulting revenue is separately assumed at INR 20,000/month, without a cost estimate. Taxes, financing, yield, clinical/regulatory work, warranty, service/calibration and working-capital timing are not fully represented. These values are scenario arithmetic, not profit or revenue achieved by AirTon.

Capital totals and debt/equity splits also conflict. Component costs summing to INR 20,000, a separate INR 5,000 variable-cost figure and INR 25,000 COGS require reconciliation. A startup budget must include landed parts, fixture/tooling, calibration labour, rejects, field service, replacements, study/regulatory work, cloud operations and clinician workflow cost. Device sale versus clinic-managed rental/loan needs attributable affordability and service-capacity research.

The recovered patent document is a drafting guide, and the 2026 patent-style drawing is a concept record. Neither proves filing, novelty or grant. Prior portable non-contact and handheld tonometer publications exist. A scoped prior-art and freedom-to-operate review needs current legal status, claim mapping, ownership, inventorship and counsel review; no legal conclusion is made here. [S38-S40]

See business reconciliation (../docs/business/baseline.md), financial scenarios (../business/financial-scenarios.json) and IP status (../docs/business/ip-status.md).

9. Next decisions and evidence gates

Priority	Work package	Reviewable exit evidence
Immediate	Review founder-selected indication, users and care responsibilities	Clinician-reviewed intended use and signed design inputs
Immediate	Obtain native CAD/PCB/Proteus, firmware, as-built BOM and bench records	Provenance/revision inventory and measured evidence
Immediate	Select and characterize a physical observable	Model-eye traces, timing/uncertainty budget and go/no-go review
Immediate	Define independent safety controls	Risk/control crosswalk and bench fault-verification criteria
Before further ML claims	Resolve dataset/checkpoint/classes and correct evaluation	Rights-cleared patient-group splits, reproducible preprocessing, reserved test and reviewed card
Before supervised study	Freeze comparator protocol and permissions	Authorized protocol, statistics, consent/data plan and trained operators
Before home study	Verify training, positioning and complete delivery/review pathway	Critical-task plan, safe failures, care assignment and service capability
Before production	Control manufacturing, calibration and operations	Reviewed master/device history records, supplier evidence and release decision
Before launch claims	Reconcile legal, clinical, business and IP states	Applicable permissions, verified claims, attributable costs and service capacity

Retinal classification and bilateral acquisition are optional expansions with their own evidence gates. They should not prevent the pressure-monitoring problem from receiving adequate metrology, home-use and service work. The dated roadmap (../docs/project/roadmap.md), backlog (../docs/project/backlog.md) and design record index (../docs/quality/design-record-index.csv) track this work.

Local software checks address fictional reference behavior and prototype build integrity. They cannot close physical, clinical, usability, manufacturing or regulatory gates. The verification record (../docs/software/verification.md) records executed checks and their limits; CI is configured, not remotely executed.

Repository maintenance should retain dated reports, immutable source versions, review states and change impact. Attach new results to requirements/risks and record actual failures as well as successful measurements. Store patient data and restricted weights in governed artifact storage, not merely ignored directories.

10. Citations and project provenance

S-identifiers refer to the full 40-source primary evidence register (../research/source-register.md). The following references support the central decisions in this report. Publication results, comparator specifications and regulator guidance do not validate AirTon.

- S01: National Eye Institute, Glaucoma and Eye Pressure.
- S02: NICE NG81, glaucoma recommendations.
- S04: Kempf et al., eye deformation in non-contact tonometry, DOI 10.1109/IEMBS.2006.259905.
- S08: iCare HOME2 patient instructions: existing home-use comparator, different technology; verify market-specific current IFU.
- S15: TRIPOD+AI reporting guidance.
- S16: Good Machine Learning Practice principles.
- S18: CDSCO ophthalmology device classification.
- S19: Medical Devices Rules, 2017, read with applicable amendments in the official Gazette index.
- S21-S22: MeitY final DPDP Rules and timeline index, Gazette text.
- S23: ISO 8612:2009 catalogue.
- S24: ISO 15004-2:2024 catalogue.
- S25: IEC 62304 catalogue, 2006+A1:2015.
- S26-S27: FHIR R4 Observation, DiagnosticReport.
- S31: Air-puff model-cornea numerical study.
- S32: ORIGA original publication, author bibliographic record.
- S33: Kaggle glaucoma-detection mirror named in notebook.
- S34-S35: ACRIMA data record, original 2019 study.
- S36: FDA, Design Considerations for Devices Intended for Home Use, final November 2014.
- S37: ISO 14971:2019 catalogue, confirmed 2025.
- S38-S40: IP India e-services, US8652044B2, US4724843A: prior-art orientation, not an FTO opinion.

Actual project sources P01-P14, connected-provider metadata and immutable revisions are in the project evidence register (../docs/project/evidence-register.md) and recovery manifest (../artifacts/recovery-manifest.json). The user-linked original DeepLearn notebook and actual React repository are central recovered source records. The supplied 38-page July deck is preserved unchanged locally. Original source documents retain their own attribution and rights; repository organization does not confer permission to redistribute or commercially use datasets, code or models.